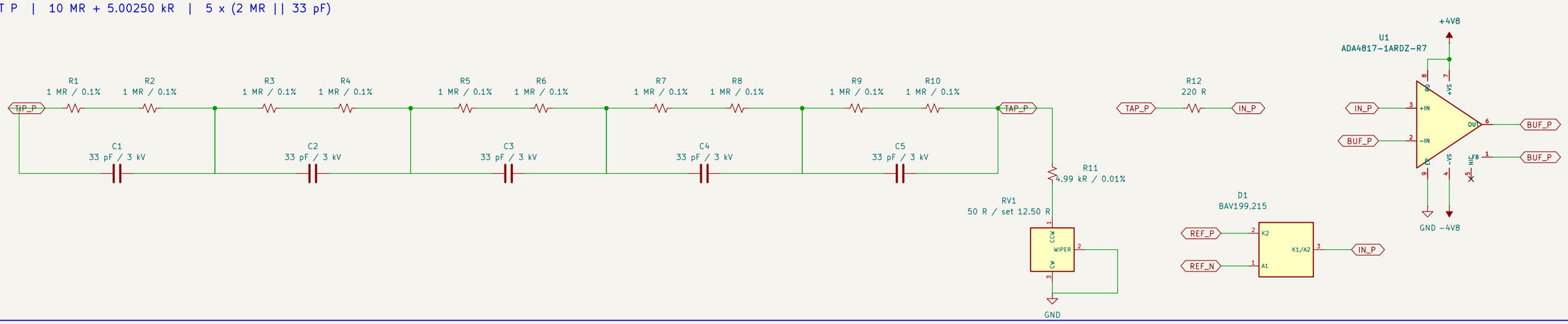
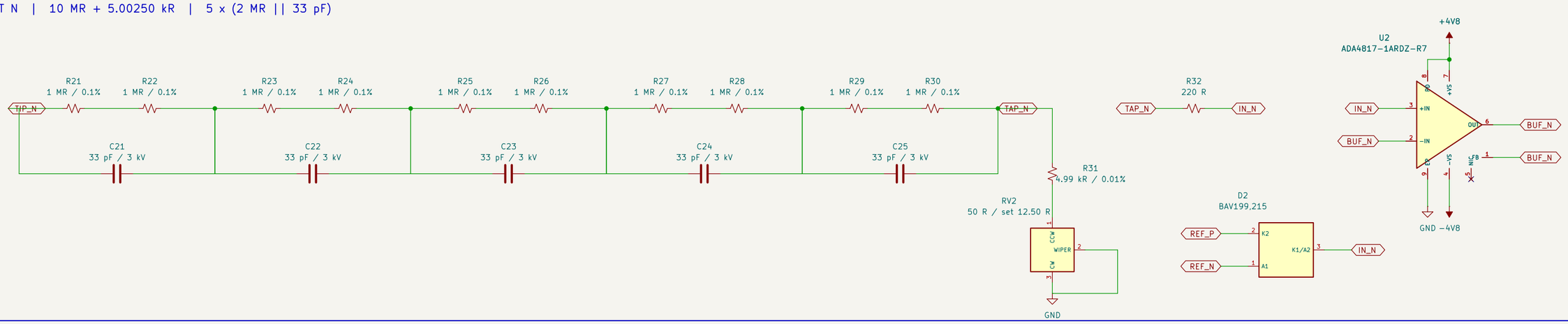
2000:1 INPUTS / matched geometry, fixed grounded shield

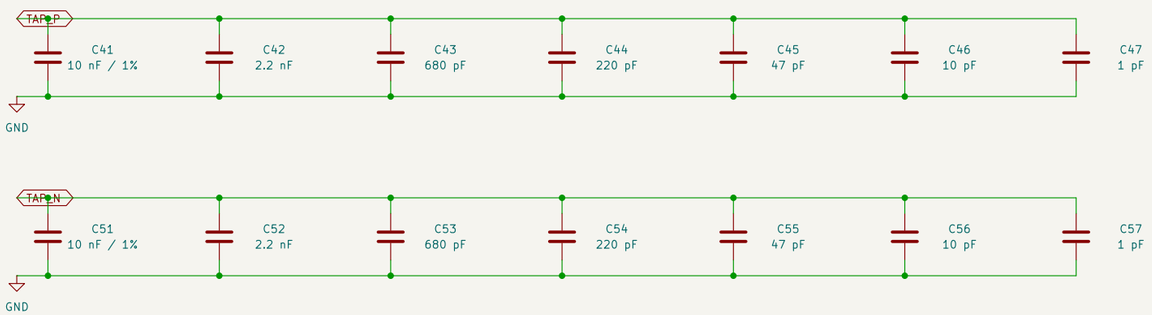
INPUT P | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)

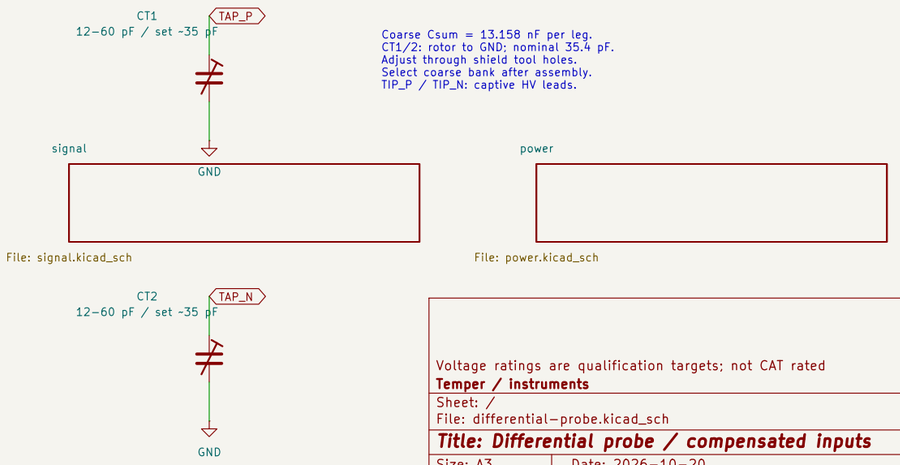


INPUT N | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)



AC COMPENSATION / coarse C0G bank + accessible fine trim





Coarse Csum = 13.158 nF per leg.
CT1/2: rotor to GND; nominal 35.4 pF.
Adjust through shield tool holes.
Select coarse bank after assembly.
TIP_P / TIP_N: captive HV leads.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /
File: differential–probe.kicad_sch

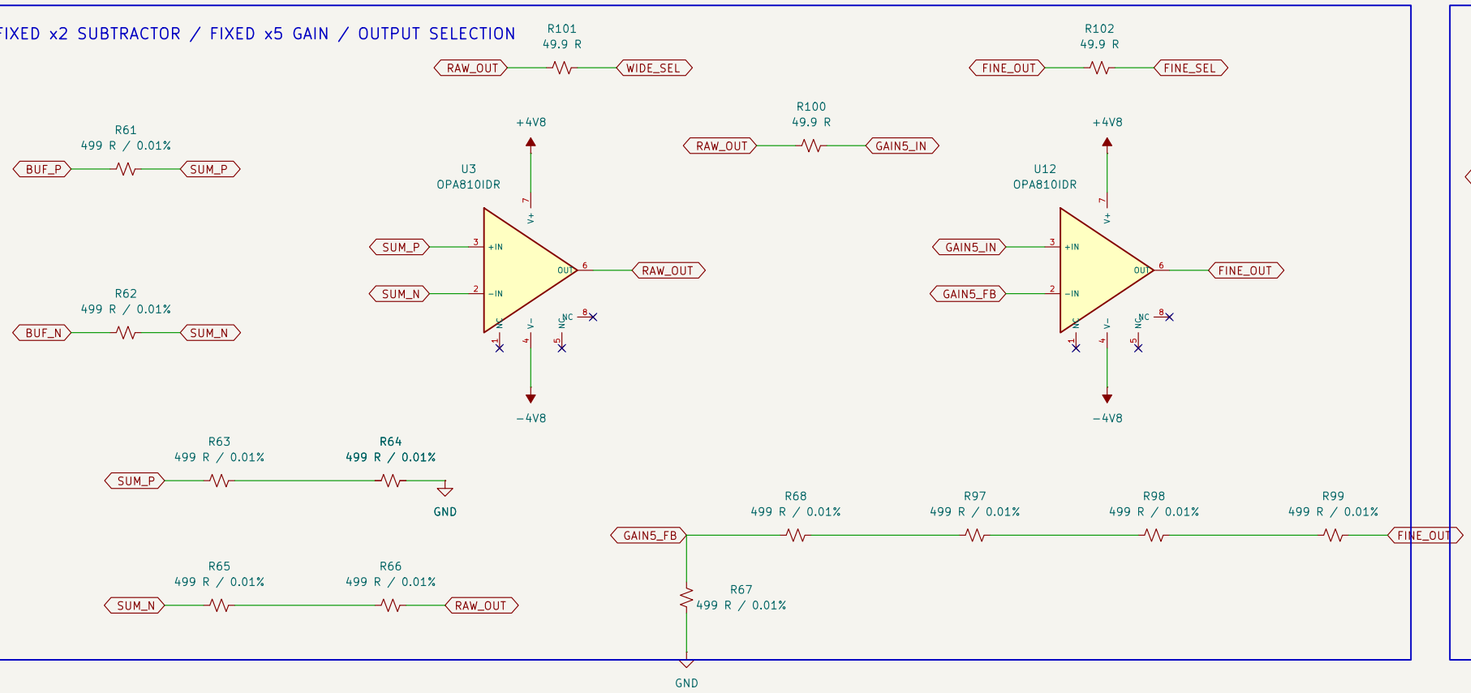
Title: Differential probe / compensated inputs

Size: A3	Date: 2026–10–20	**Rev: B prototype**
KiCad E.D.A. 10.0.4		Id: 1/3

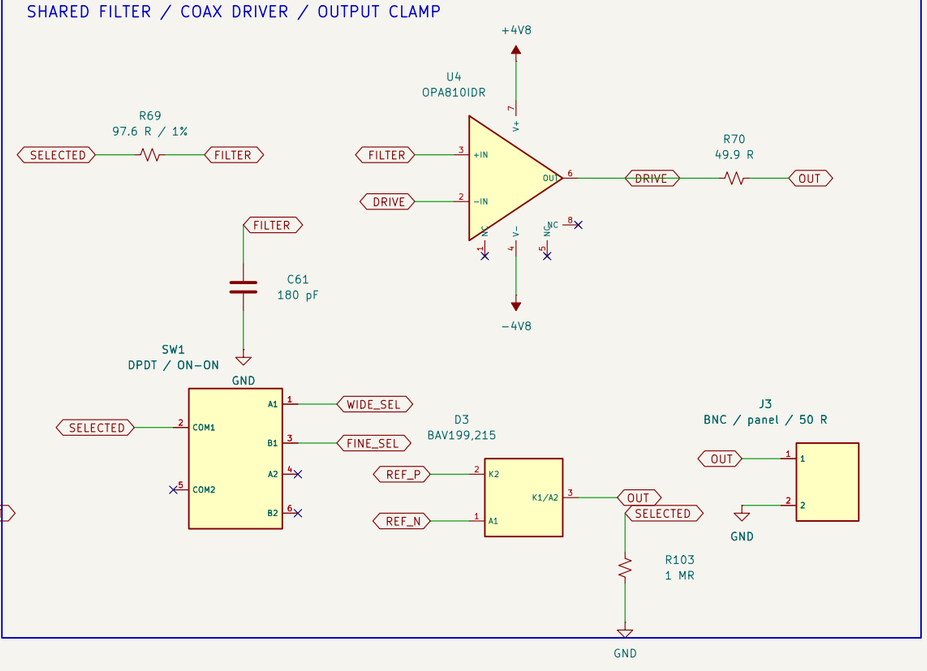
1000:1 / 200:1 OUTPUT / scope input = 1 MR

SW1 pin 2 selects 1 = 1000:1, 3 = 200:1. No switched feedback. Verify BBM contacts; allow settling during changes.

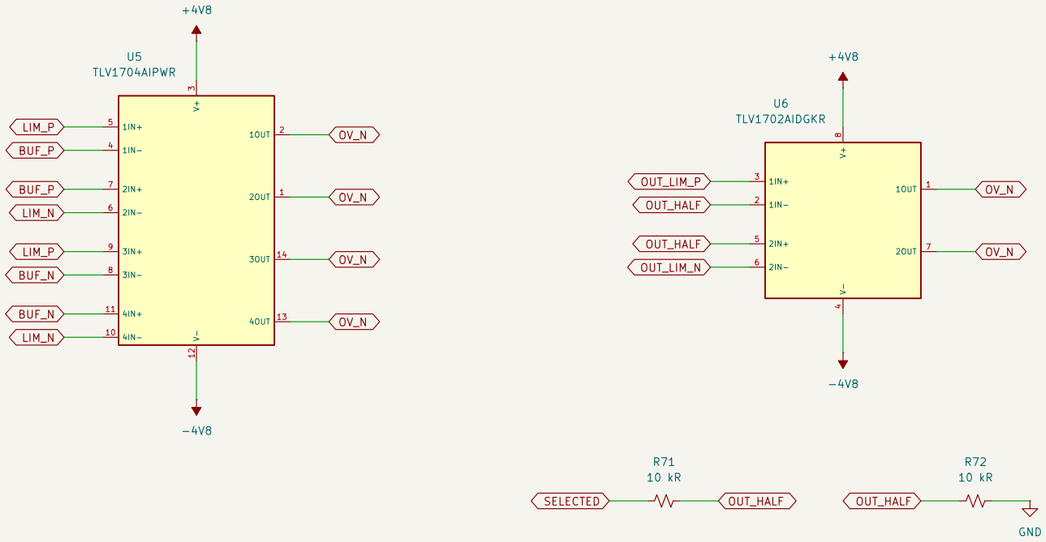
FIXED x2 SUBTRACTOR / FIXED x5 GAIN / OUTPUT SELECTION



SHARED FILTER / COAX DRIVER / OUTPUT CLAMP

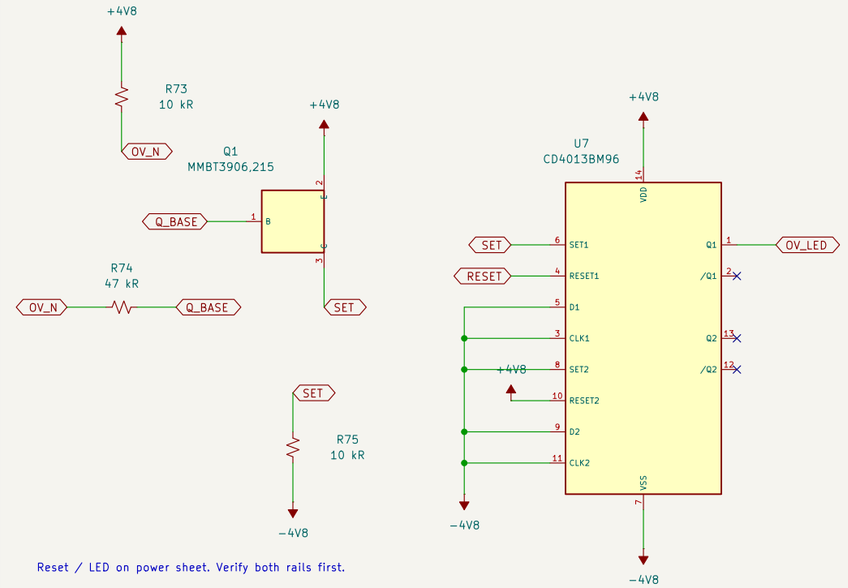


OPERATING-LIMIT WARNING / tips ~1406 V; selected signal ~1.266 V



TLV170x: 560 ns typical delay. Indicator is not a fast peak detector.
Unused x5 stage may saturate in 1000:1 range; selected signal alone drives output warning.

LATCHED OVERLOAD / RESET



Reset / LED on power sheet. Verify both rails first.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

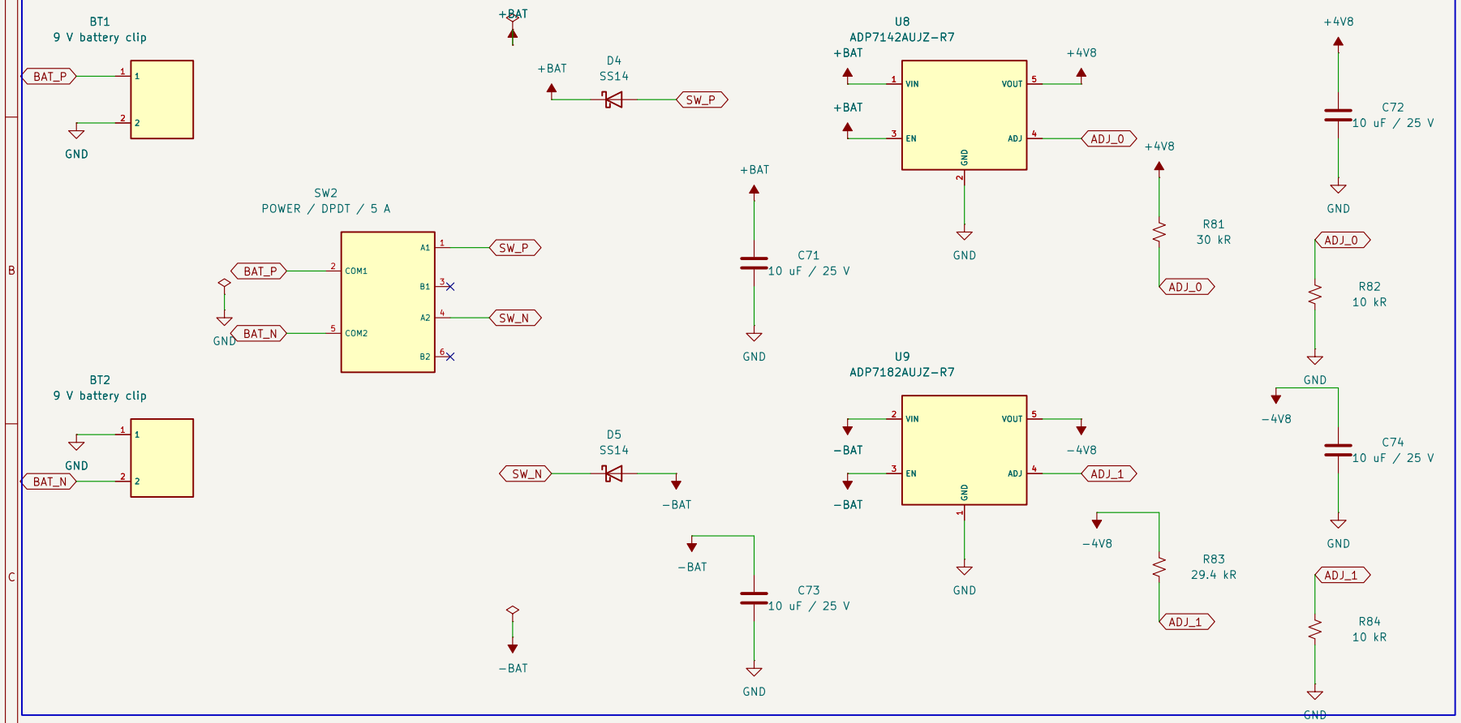
Sheet: /signal/
File: signal.kicad_sch

Title: Differential probe / gain, output and overload

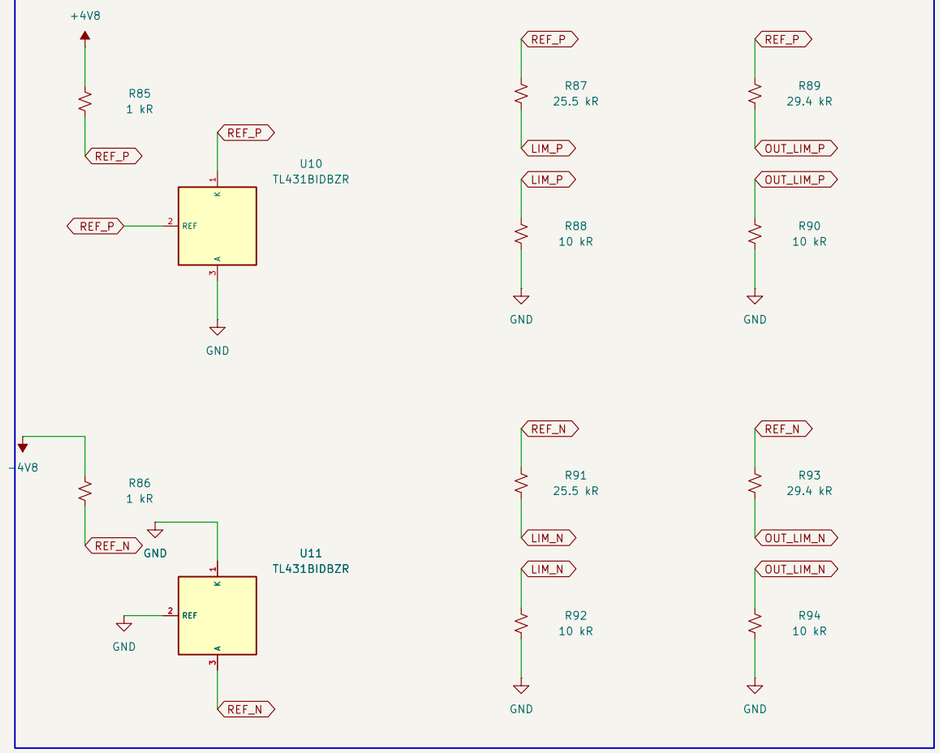
Size: A3	Date: 2026-10-20	**Rev: B prototype**
KiCad E.D.A. 10.0.4		Id: 2/3

BATTERY POWER / scope shield defines circuit ground

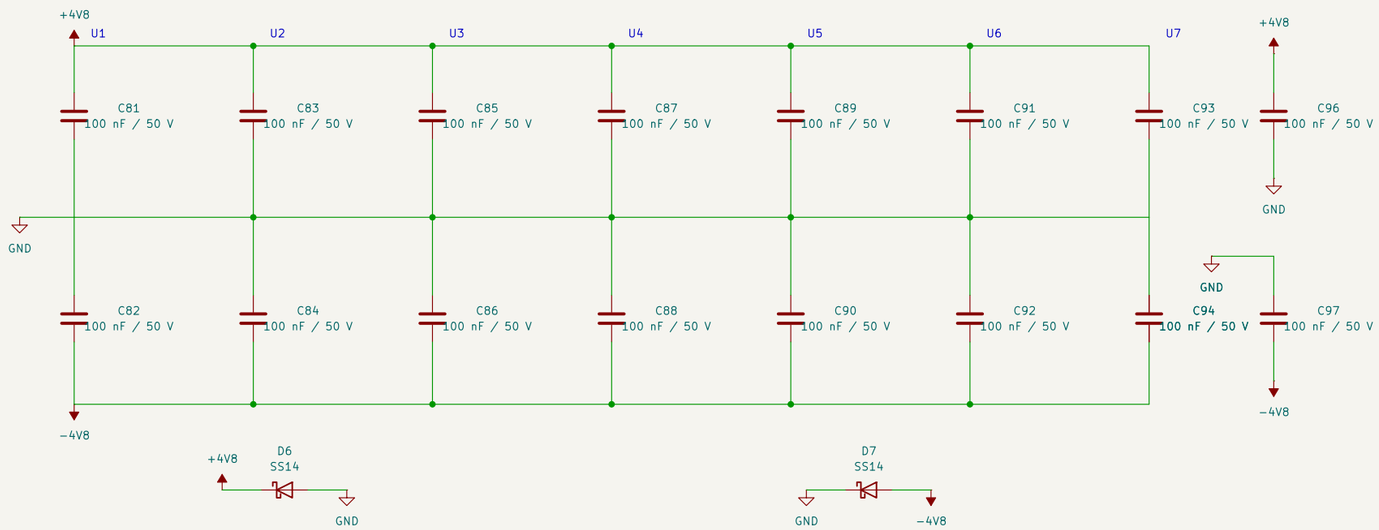
TWO 9 V BATTERIES / +/−4.8 V LOW-NOISE RAILS



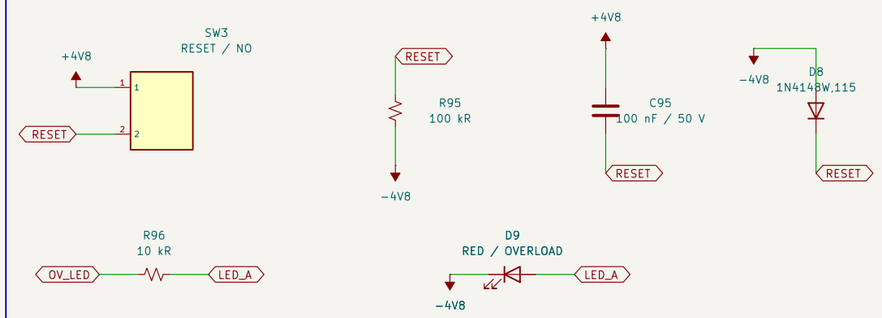
SHUNT CLAMPS / COMPARATOR THRESHOLDS



LOCAL DECOUPLING / RAIL REVERSAL DIODES



RESET / STATUS / POWER VALIDATION



Two 9 V alkaline batteries; center tap = GND.
Verify rails, reset, then connect DUT.
No charging / USB connection during measurement.

Voltage ratings are qualification targets; not CAT rated

Temper / instruments

Sheet: /power/
File: power.kicad_sch

Title: Differential probe / battery supply and references

Size: A3	Date: 2026-10-20	**Rev: B prototype**
KiCad E.D.A. 10.0.4		Id: 3/3